Introduction to Feedback Control – General Control Design Principles

(反馈控制介绍 – 一般控制设计原理)

Why Feedback ?

- **Objectives of Feedback (反馈的目的):**
  - **Stabilize unstable systems (使不稳定系统稳定)**
    - Adjust the closed-loop system's dynamic behavior (调节闭环系统的动态行为). For example, unstable plants such as inverted pendulum and the position control of DC motor (如倒立摆系统和直流电机的位置控制等不稳定系统).
  - **Improve system performance to meet stringent performance requirements (提高系统性能以满足严格的性能要求)**
    - Steady State Performance (稳态性能): for example, reduce steady state error due to disturbances ... (如降低干扰引起的稳态误差)
    - Transient Performance (瞬态性能): for example, reduce rise time and settling time to speed up system response , … (如降低上升时间和调整时间以加快系统响应)
  - **Reduce the effect of modeling uncertainty and various disturbances (降低模型不确定和各种干扰的影响)**
    - Employing high gain feedback at the right location and frequency range (在适当的位置和频段采用高增益反馈)
    - Possible instability of closed-loop system in the presence of various physical constraints (由于各种物理限制可能导致闭环系统不稳定)

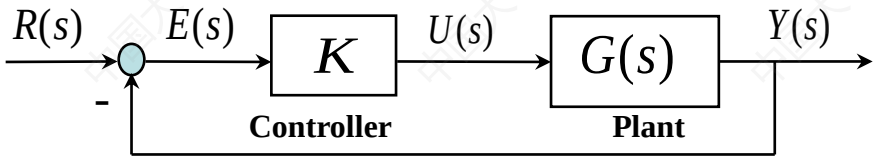
Why Feedforwad ?

- **Model Compensation in Advanced Nonlinear Controls (先进非线性控制中的模型补偿)**
  - Makes the control input close to the desired control input that is needed to accomplish the task (使控制量接近满足任务所需要的期望控制量)
  - Only small amount of control correction needs to be provided by feedback control action, which results in smaller tracking error (keep in mind that feedback control needs tracking error to generate the control action) (使反馈控制仅提供需要的小幅控制量修正，以获得较小的跟踪误差(反馈控制需要跟踪误差来产生控制行为))

- **Applications of Feedforward (前馈的应用)**
  - For applications having stringent performance requirements such as control of precision electro-mechanical devices (有严格性能需求的应用，如电机系统的高精度控制)
  - Usefulness of feedforward heavily depends on the accuracy of models used for physical plants, which normally have quite large variations of system parameters (前馈的有效性严重依赖于参数可能有较大变化的物理系统的模型精度)

General Control Design Principles

- **Control design is nothing but an *Inversion Process* (控制设计除了过程求逆没有什么)**
- **The inversion can be achieved by two key mechanisms (过程求逆可通过两种方式获得):**
  - **(High-Gain) Feedback and (高增益反馈)**
  - **(Model Compensation) Feedforward (模型补偿前馈)**



$$\frac{U(s)}{R(s)} = \frac{K}{1+KG} \overset{\text{large } K}{\approx} G^{-1}$$

General Control Design Principles

- **High-Gain Feedback (高增益反馈)**
  - Gives approximate inversion in the presence of modeling errors and disturbances (在建模误差和干扰存在时给出近似逆)
  - In practice, the choice of feedback gain is part of a complex web of design trade-offs (实际上反馈增益的选择是设计矛盾复杂性的一部分)

high-gain (高增益) ⇒ **high sensitivity to measurement noises (对测量噪声具有高敏感性)** ⇒ **makes the stability of closed-loop system sensitive to control input saturation and neglected high-frequency dynamics (使闭环系统的稳定性对控制输出饱和忽略的高频动力学较敏感)**

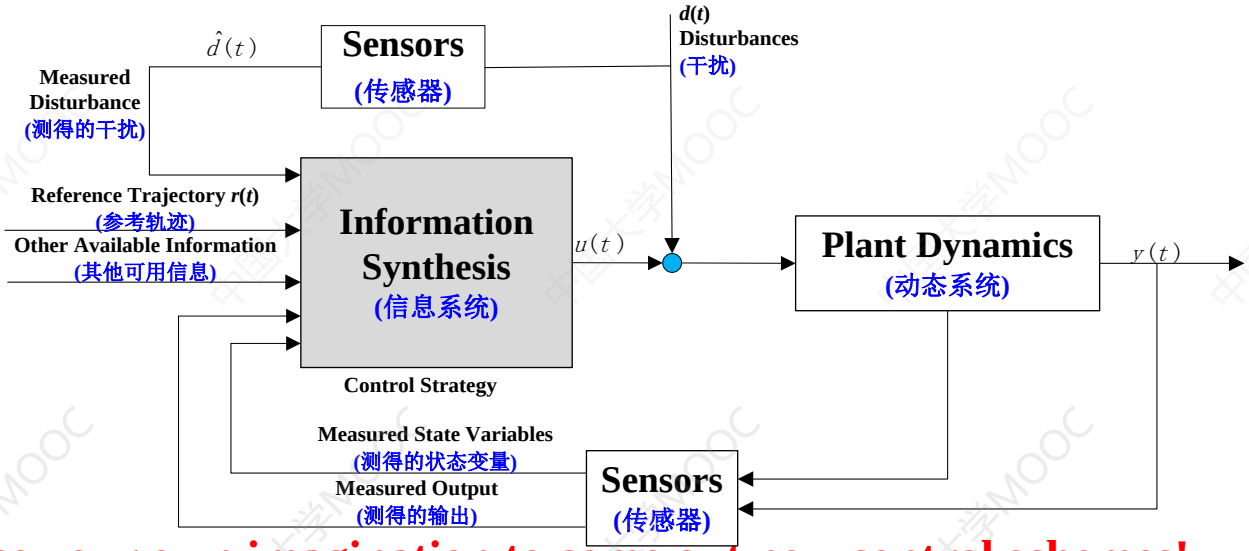
  - Understanding and balancing these trade-offs is the essence of feedback control design (理解和平衡这些设计矛盾是反馈控制设计的本质)

Use nonlinear feedback instead of linear feedback to achieve a better trade-off ! (用非线性反馈来代替线性反馈以更好的平衡矛盾)

- **On-line learning (在线学习)**
  - A key to have a good model compensation of feedforward design (是前馈设计获得好的模型补偿的关键)

General Structure of Controller

In general, a controller is nothing but a strategy to determine a control action based on all available information; information not only comes from the measured output but also from the measured internal state variables, measured disturbance, reference trajectory, and plant model structure (通常，控制器就是基于所有可得信息来决定一个控制行为的策略，这些信息不仅来自于测量输出，也来自于测量内部状态，测量干扰，参考轨迹和模型结构)



So use your own imagination to come out new control schemes!

Summary

- **Objectives of Feedback and Feedforward (反馈和前馈控制的目的)**
- **General Control Design Principles (一般控制设计原理)**
  - **Process inverse: High-gain Control, Model Compensation by Feedforward (过程求逆:高增益反馈和基于前馈的模型补偿)**
- **General Structure of Controller (控制器的一般结构)**